Banach space examples and necessity of assumptions

We now turn to examples involving non-normal operators or failure of commutativity. These will illustrate why conditions like the Ritt resolvent bound and commutativity of layers are necessary for convergence in general.

Example 4.4 (Ritt operator on ℓ^2 with spectral radius 1)**.** We give a simple finite-dimensional example to illustrate Theorem C. Let $A = \begin{pmatrix} 1 & 0 \\ 0 & r \end{pmatrix}$ acting on $\mathbb{C}^2$, where $0 < r < 1$. This A is a diagonal matrix with $\sigma(A) = \{1, r\}$; it is power-bounded (even contractive) and clearly satisfies a Ritt resolvent condition (being diagonalizable). Now take $K = 1$ (a single layer), and let $\varphi_1(z) = z$ (the trivial transform). Then $\Psi(A) = A$. We see that $A^n = \begin{pmatrix} 1 & 0 \\ 0 & r^n \end{pmatrix}$, which converges as $n \to \infty$ to $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. This is exactly the projection onto the eigenspace at 1, as expected by the general theory (in fact, the Katznelson–Tzafriri theorem directly applies).

A more interesting example is obtained by taking $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ acting on $\mathbb{C}^2$. This A is not a contraction, but has spectral radius 1 and is a Ritt operator with resolvent bound (since $\sigma(A) = \{-1, 1\}$, and indeed A is diagonalizable so it admits a bounded functional calculus). Now choose $\varphi_1(z) = z^2$. Then $\varphi_1(1) = 1$ and $\varphi_1(-1) = 1$ as well. The iteration $A^{(n)} = (\varphi_1 \circ \cdots \circ \varphi_1)(A) = (\varphi_1(A))^n$ satisfies $\varphi_1(A) = A^2 = I$. Thus $A^{(n)} = I$ for all n, which trivially converges (to I). In this case $\Sigma_{\text{fix}} = \{1, -1\}$ is the whole spectrum, so the final projection is I. Note that if we had not included -1 as a fixed point of φ_1, the sequence would not converge. For instance, had we taken $\varphi_1(z) = \frac{1+z}{2}$, then $\varphi_1(1) = 1$ but $\varphi_1(-1) = 0$. In that hypothetical scenario, $\varphi_1(A) = \frac{1}{2}(I + A) = \frac{1}{2}\begin{pmatrix} 2 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, whose powers converge to $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ (the projection onto the eigenspace for eigenvalue 1). This is indeed $E_A(\{1\})$, consistent with $\Sigma_{\text{fix}} = \{1\}$ in that scenario.

Finally, we mention that if A is power-bounded but does not have an actual eigenvalue at 1 (only a spectral value without eigenvector), then even if $\varphi_k(1) = 1$, one cannot expect powers to converge without additional resolvent assumptions. A typical example is a weighted shift on ℓ^p that is power-bounded but not uniformly. This is known to be non-mean-ergodic (the Cesàro averages do not converge to a projection in strong operator topology) if it has peripheral point spectrum but no eigenvalue at 1. Such an example shows the necessity of the Arendt–Batty spectral condition in Theorem C part (b). See e.g. [19, 20] for details.

Example 4.5 (Non-commuting layers and cycles)**.** The commutativity of the layers $\Phi_k(A)$ (ensured in our setting by being functions of a common A) is crucial for the process to converge. If the layers do not commute, the iteration can cycle. For a concrete 2×2 example, consider the matrices on $\mathbb{C}^2$, where S is the swap matrix $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. These are two layers that do not commute with each other in general. Now let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ which is a non-normal operator (a Jordan block with eigenvalue 1). Then $SAS = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ and $S(SAS)S = A$. Thus the iteration cycles with period 2 and never converges (it oscillates between A and SAS). This highlights that allowing non-commuting layers can destroy convergence [12, 13], as mentioned in the introduction.

We note that even though A was power-bounded and mean ergodic since it is of class C_1. with $\sigma(A) = \{1\}$, the process of alternating A with its conjugate by S leads to a cycle. This is reminiscent

of the alternating projection method in convex optimization (where the composition of two non-commuting orthogonal projections need not converge in norm, but strongly it does by classical results—here we had $\|A \cdot (SAS)\| = 1$ always, a neutral case, and indeed only weak convergence would hold even if some conditions are met, or in our finite case it cycles exactly).

Appendix A: Two basic tools for the normal functional calculus

Lemma .6 (Dominated convergence for the normal functional calculus)**.** *Let A be a normal operator on a complex Hilbert space H with spectral measure E_A. Let $(g_n)_{n\geq 1}$ be bounded Borel functions on $\sigma(A)$ such that $\sup_n \|g_n\|_\infty \leq M < \infty$ and $g_n(\lambda) \to g(\lambda)$ pointwise on $\sigma(A)$. Then $g_n(A) \xrightarrow{SOT} g(A)$ as $n \to \infty$.*

Proof. For each $x \in H$, define a complex measure $\mu_x(B) := \langle E_A(B)x, x\rangle$, which is a finite measure supported on $\sigma(A)$. By the Borel functional calculus, one has $\|(g_n(A) - g(A))x\|^2 = \int_{\sigma(A)} |g_n(\lambda) - g(\lambda)|^2 \, d\mu_x(\lambda)$. Now since $|g_n - g|^2 \leq 4M^2$ (bounded by some integrable constant) and $|g_n(\lambda) - g(\lambda)|^2 \to 0$ for each λ, we can apply the dominated convergence theorem to conclude that $\int |g_n - g|^2 \, d\mu_x \to 0$. Thus $\|(g_n(A) - g(A))x\| \to 0$ for every x, i.e. $g_n(A) \to g(A)$ in the strong operator topology. □

Lemma .7 (Composition in the normal functional calculus)**.** *Let A be normal, h a bounded Borel function on $\sigma(A)$, and g a bounded Borel function on $h(\sigma(A))$. Then $g(h(A)) = (g \circ h)(A)$.*

Proof. We prove this first for simple functions. Suppose $h(\lambda) = \sum_{i=1}^N c_i 1_{B_i}(\lambda)$, where B_i are pairwise disjoint Borel sets. Then $h(A) = \sum_{i=1}^N c_i E_A(B_i)$. Now define $f := g \circ h$. Then $f(\lambda) = \sum_{i=1}^N g(c_i) 1_{B_i}(\lambda)$ is another simple function, and $f(A) = \sum_{i=1}^N g(c_i) E_A(B_i)$. On the other hand, $g(h(A))$ can be computed as $g(\sum_{i=1}^N c_i E_A(B_i)) = \sum_{i=1}^N g(c_i) E_A(B_i)$ since the $E_A(B_i)$'s are pairwise orthogonal projections and g is evaluated scalar-wise on the diagonal form of $h(A)$. Thus indeed $g(h(A)) = f(A)$.

For a general h and g, we can approximate h by simple functions h_n such that $h_n(\lambda) \to h(\lambda)$ uniformly on $\sigma(A)$, and similarly approximate g by simple functions g_n uniformly on the range $h(\sigma(A))$. Then by continuity of the functional calculus (using Lemma .6), we have $g(h(A)) = \lim_{n\to\infty} g_n(h_n(A)) = \lim_{n\to\infty}(g_n \circ h_n)(A)$. Since $g_n \circ h_n \to g \circ h$ uniformly on $\sigma(A)$, the latter equals $(g \circ h)(A)$. □

Appendix B: Two-point Schur/Nevanlinna–Pick lemma and interpolation

B.1 Reminder: the first Schur step

If R is a Schur function (holomorphic $R : \mathbb{D} \to \overline{\mathbb{D}}$) with $R(0) = 0$, then there exists a Schur function Φ such that $R(z) = z\Phi(z)$. This is the first step of the Schur algorithm: the function Φ is again Schur and is unique if one additionally prescribes $\Phi(0) = R'(0)$. Conversely, any function of the form $z\Phi(z)$ is Schur whenever Φ is Schur.

B.2 Two-point Schur/Nevanlinna–Pick interpolation with one interior and one boundary node

Fix $t \in (0, 1)$ and consider the interpolation problem in the Schur class $\mathcal{S} := \{s : \mathbb{D} \to \overline{\mathbb{D}} \text{ holomorphic}\}$: $s(t) = 0$ and $s(1) = 1$. Let $b_t(z) = \frac{z-t}{1-tz}$ denote the disk automorphism with $b_t(t) = 0$ and $b_t(1) = 1$.

Lemma .8 (Two-point Schur/Nevanlinna–Pick)**.** *The set of all Schur solutions to the above interpolation problem is* $\{s(z) = b_t(z)\Phi(b_t(z)) : \Phi \in \mathcal{S}, \Phi(1) = 1\}$*. In particular, the minimal-degree inner solution is the Blaschke factor* $s^*(z) = b_t(z)$*.*

Proof. If s is any Schur solution, define $\Psi(w) := s(b_t^{-1}(w))$ for w in the disk. Then Ψ is a Schur function with $\Psi(0) = s(t) = 0$ and $\lim_{w\to 1} \Psi(w) = s(1) = 1$. By the Schur algorithm applied to Ψ (see B.1 above), we have $\Psi(w) = w\,\Phi(w)$ for some Schur Φ. Moreover, since Ψ tends to 1 at the boundary point 1, a standard result (Carathéodory's lemma, or a special case of Julia's lemma) implies that Φ also tends to 1 nontangentially at 1. Converting back to s, we get $s(z) = b_t(z)\Phi(b_t(z))$ with $\Phi \in \mathcal{S}$ and $\lim_{z\to 1} \Phi(z) = 1$. This shows that any solution has the desired form. Conversely, if $s(z) = b_t(z)\Phi(b_t(z))$ with Φ as stated, then obviously $s(t) = 0$, and since $\Phi(1) = 1$ we have $\lim_{z\to 1} s(z) = (\lim_{z\to 1} b_t(z))(\lim_{z\to 1} \Phi(b_t(z))) = 1 \cdot 1 = 1$. So s is indeed a solution. The uniqueness of the minimal-degree (finite Blaschke product) solution follows from Schur–Nevanlinna–Pick theory, but it is clear here that if Φ is inner, then so is s, and the simplest inner Φ with $\Phi(1) = 1$ is the constant 1 function. Thus the minimal inner s is $b_t(z) \cdot 1 = b_t(z)$ itself. □

Appendix C — Fejér-type stabilization, boundary separation, and a normalized ω-example

This appendix supplies (i) a rigorous Fejér-type stabilization bound $\alpha^* \le \omega$ in the normal case, (ii) a short lemma ensuring power-boundedness of $\Psi(A)$ under the standing H^∞-calculus bound in the Banach case, (iii) a precise boundary–separation statement implying the joint fixed-point identification used with Lemma 3.3, and (iv) a corrected two-layer example (a normalized version of Example 4.1) that attains the extremal countable stage $\alpha^* = \omega$. These items close the points raised around Lemma 2.6, Theorem 3.1, and Examples 4.1/4.5 without introducing new hypotheses beyond those already adopted in the paper. References to the main text are indicated inline.

C.1 Fejér-type stabilization bound $\alpha^* \le \omega$ (normal case)

We work under the "normal-case" standing assumptions of Section 2: A normal, p_k holomorphic on a neighborhood of $\sigma(A)$ with $\sup_{\lambda\in\sigma(A)} |p_k(\lambda)| \le 1$, $p_k(1) = 1$, and the peripheral fixed-point property. Write $f := p_K \circ \cdots \circ p_1$ and $f^m := \underbrace{f \circ \cdots \circ f}_{m \text{ times}}$. The spectral-calculus facts used below are precisely Lemma .6 (dominated convergence) and Lemma .7 (composition) in Appendix A of the manuscript.

Lemma .9 (Idempotence of the pointwise limit)**.** *Let* $\chi : \sigma(A) \to \{0, 1\}$ *be the pointwise limit* $\chi(\lambda) = \lim_{m\to\infty} f^m(\lambda)$ *given by Lemma 2.6. Then* $f \circ \chi = \chi$ *pointwise on* $\sigma(A)$*.*

Proof. By Lemma 2.6 there are exactly two possibilities: either the Denjoy–Wolff point of f lies in $\mathbb{D}$ and equals 0, in which case $f(0) = 0$ and $f^m(\lambda) \to 0$ for all $\lambda \in \sigma(A)$; or the Denjoy–Wolff point is the boundary point 1, in which case $f(1) = 1$ and $f^m(\lambda) \to 1$ for all λ in the peripheral fixed set (and to 0 otherwise, as established in Lemma 2.6). In either case the only limit values are the fixed points 0 and 1, hence $f(\chi(\lambda)) = \chi(\lambda)$. □

Proposition .10 (Stabilization at the first limit stage)**.** *Under the normal-case assumptions, the ordinal iteration stabilizes at the first limit stage:* $A^{(\omega)} = \chi(A)$ *and* $A^{(\omega+\beta)} = \chi(A)$ *for all ordinals* $\beta \ge 1$*. In particular,* $\alpha^* \le \omega$*.*